

Java Assignment 2

Instructions

- Use Scanner for input wherever required.
 - Use clear prompts and meaningful variable names.
 - Display output in a user-friendly format.
 - Add comments to explain the logic.
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1. Electricity Bill Calculator

Create a Java program to calculate a household electricity bill.

Requirements

- Accept customer name, consumer number, and units consumed.
 - Calculate bill using slab rates such as:
 - First 100 units → Rs. 5 per unit
 - Next 100 units → Rs. 7 per unit
 - Above 200 units → Rs. 10 per unit
 - If the bill exceeds Rs. 1500, apply a 5% surcharge.
 - Display complete bill details.
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2. Student Result Processing System

Create a program for a school to manage a student's marks.

Requirements

- Accept student name, roll number, and marks in 5 subjects.
- Calculate total, average, percentage.
- Display grade using conditions:
 - 90 and above → A
 - 75 to 89 → B
 - 60 to 74 → C
 - 40 to 59 → D
 - Below 40 → Fail

- Show whether the student is promoted or not.
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3. ATM Simulation

Create a simple ATM system.

Requirements

- Show a menu using switch-case:
 1. Check Balance
 2. Deposit Money
 3. Withdraw Money
 4. Exit
 - Start with an initial balance.
 - Prevent withdrawal if balance is insufficient.
 - Continue showing the menu until the user chooses Exit.
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4. Grocery Store Bill Generator

Create a billing system for a small grocery shop.

Requirements

- Accept item names, quantity, and price for multiple items.
 - Store item totals in arrays.
 - Calculate total bill amount.
 - Apply discount:
 - 10% discount if bill is above Rs. 3000
 - Print the final bill in proper format.
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5. Employee Salary Slip Generator

Create a Java program to generate salary slips for employees.

Requirements

- Accept employee name, ID, basic salary.

- Calculate:
 - HRA = 20% of basic salary
 - DA = 10% of basic salary
 - PF = 8% of basic salary
 - Calculate net salary.
 - Display a formatted salary slip.
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6. Bus Ticket Booking System

Create a program for booking bus seats.

Requirements

- Assume a bus has 10 seats.
 - Use an array to mark seats as booked or available.
 - Show a menu:
 1. View Available Seats
 2. Book a Seat
 3. Cancel a Seat
 4. Exit
 - Do not allow double booking.
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7. Mobile Recharge System

Create a recharge application for prepaid mobile users.

Requirements

- Ask for mobile number and recharge amount.
 - Show recharge plans using switch-case.
 - Example:
 - Rs. 199 → 28 days validity
 - Rs. 399 → 56 days validity
 - Rs. 599 → 84 days validity
 - Print recharge confirmation with plan details.
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8. Hospital Patient Record System

Create a simple patient management program.

Requirements

- Create a class Patient with:
 - patient ID
 - patient name
 - age
 - disease
 - Create objects for at least 3 patients.
 - Display patient details.
 - Add a method to check whether the patient is a senior citizen (age 60 or above).
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9. Daily Expense Tracker

Create a program to track a person's daily expenses.

Requirements

- Accept expenses for 7 days in an array.
 - Calculate:
 - total weekly expense
 - average expense
 - highest expense
 - lowest expense
 - Display the day on which the highest expense occurred.
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10. Restaurant Order Management System

Create a program for restaurant management.

Requirements

- Display a food menu with price list.
- Let the user order multiple items.
- Use loop and switch-case for repeated ordering.
- Calculate total amount.

- Add GST of 5%.
 - Print final bill.
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11. Calculator System (Overloaded Methods)

Create a Java program for a calculator:

Requirements:

- Create methods named calculate
 - Overload them for:
 - int + int
 - double + double
 - int + int + int
 - Call all methods from main
 - Display results clearly
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12. Student Counter System

Create a class Student:

Requirements:

- Instance variables: name, rollNo
- Static variable: totalStudents
- Increment totalStudents every time object is created
- Display total number of students

Bonus: Add a static method to display count

13. Bank Account System

Create a class BankAccount:

Requirements:

- Variables: accountNumber, name, balance
- Use:

- Default constructor
 - Parameterized constructor
 - Display account details
 - Create multiple objects using different constructors
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14. Salary Conversion System

Create a program:

Requirements:

- Take the salary as double
 - Convert it into:
 - int (explicit casting)
 - float (implicit casting)
 - Show the precision loss clearly
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15. Student Marks Table

Create a program:

Requirements:

- Store marks of 3 students in 3 subjects using 2D array
 - Calculate:
 - Total per student
 - Average marks
 - Display in table format
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16. Weekly Work Hours

Create a program:

Requirements:

- Each employee works different number of days
- Use jagged array to store hours
- Display all employee work hours

17. Scope Demonstration Program

Create a program:

Requirements:

- Declare variable inside:
 - Method
 - Loop
 - Block
 - Try accessing outside scope (show error via comments)
 - Print valid usages
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18. Expression Evaluator

Create a program:

Requirements:

- Take 3 numbers
 - Perform:
 - $(a + b) * c$
 - $a > b \ \&\& \ b > c$
 - $a \% b$
 - Show results with explanation
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19. Employee ID Generator

Create a class Employee:

Requirements:

- Static variable: counter
 - Each employee gets auto-incremented ID
 - Use constructor to assign ID
 - Display employee details
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20. Library System

Create two classes:

Class 1: Book

- bookId, bookName

Class 2: Library

- Method to display books

Requirements:

- Create multiple books
- Pass objects to library
- Display all books